

Bed Lighting

Monday, August 10, 2026 1:18 PM

This one took me a while. Probably the most involved one I've done so far but that's because I thought I could come off the +12V lead from the top center third brake light - which didn't work out. I found the circuit for that and even made my own harness tie in for the lead. Turns out, either I don't have enough current flowing that line to power the bed lights AND the center light and/or my bed lights just naturally need more current. I never ended up measuring the current they really needed and what was in the line.

I really just wanted the ability to have them come on with the center lights and when doors/tailgate was opened as well as be switched from that inside switch. I still got them wired up though and it's through the battery like most do.

Parts/tools I used:

Matt Gecko Bed Light + Under Hood Light kit (<https://www.tundras.com/out/?url=https%3A%2F%2Fmattgeckoleds.com%2Fshop%2Fols%2Fproducts%2Fled-bed-light-kit-with-bed-mounted-switch-toyota-tacoma-2005-2021%2Fv%2FLED-BED-LGH-KIT-201-2029&inner=TWf0dCUyMEdlY2tvJTlwQmVkJTlwTGlnaHQIMjAlMklIMjBVbmRlciUyMEhvb2QIMjBMaWdodCUyMGtpdA%3D%3D&post=3887124>) - \$151.50 total shipped (see my under hood light writeup for more info)

\$10 RED WOLF Wire Harness Cloth Tape 1" x 50 FT Electric Wiring Loom Harness Flexibility High Abrasion Resistance Adhesive for Automobile Electrical Engine Noise Dampening Insulation 2 Rolls

Wire Harness Cloth Tape - \$9.99

Add-a-circuit fuse tap (<https://www.tundras.com/out/?url=https%3A%2F%2Fwww.amazon.com%2Fdp%2FBOBW3SXFDR&inner=V2lyZSUyMEhhcm5lc3MlMjBDbG90aCUyMFRhcGU%3D&post=3887124&extra=relproding>) - \$6.79

Zip ties - should have enough in the kit from Matt Gecko

Tools:

A couple beers

Wire strippers

Solder/butt connectors/some means of connecting wire to each other

Multimeter

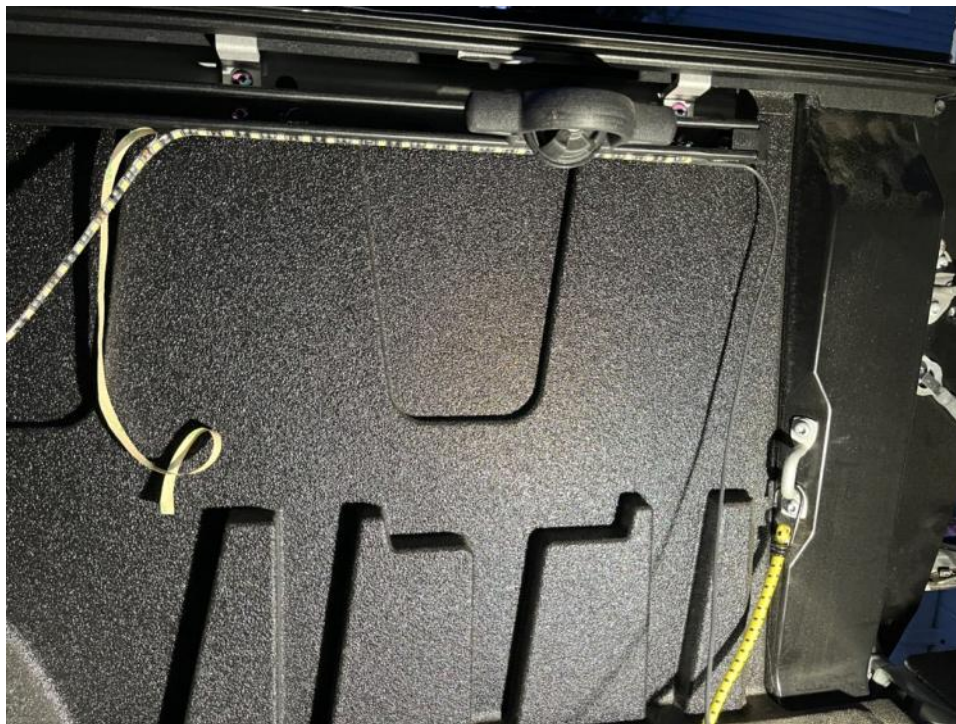
Something to fish with (I use a landscape flag)

About half-day to full-day of time (what it took me from all the business of trying to tie into the center light - should be quicker if you just go by my writeup)

I fished the positive +12V wire from the bed area back to the fuse box near the battery. You could do it the other way around if you wanted.

The kit gives plenty of wire for this. I ended up making two wire runs bonded together about half the distance (one for +12V and other for ground). Then took a single wire past that to under the hood. The ground wire stopped at an OEM ground bolt at the frame (I'll show that later).

Start by tapping on the lights around the bottom of the frame rails. I started on the passenger side where the wire connection would be and walked it around the bed. I tried centering up the lights as best I could on the rails.



I wrapped the harness run with my harness cloth tape. I found a good spot to take from the truck bed to cavity of the truck (passenger side). There's a little hole around the bottom of rail/top off the side bed where the rear pillar of the truck bed is.



Then I took the tail light off and laid to the side to access this cavity area.



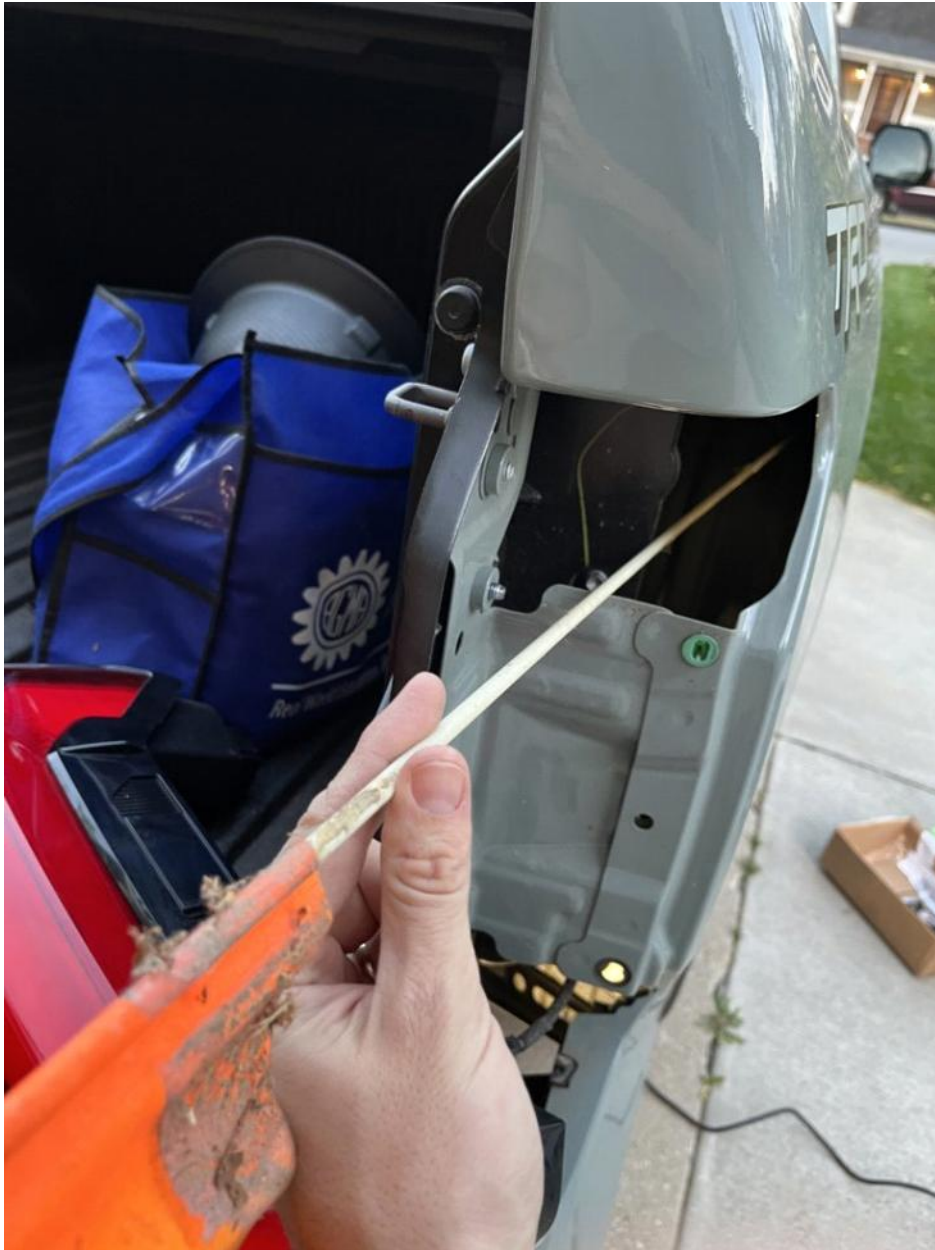
You can see the green of the flag where this pokes through.



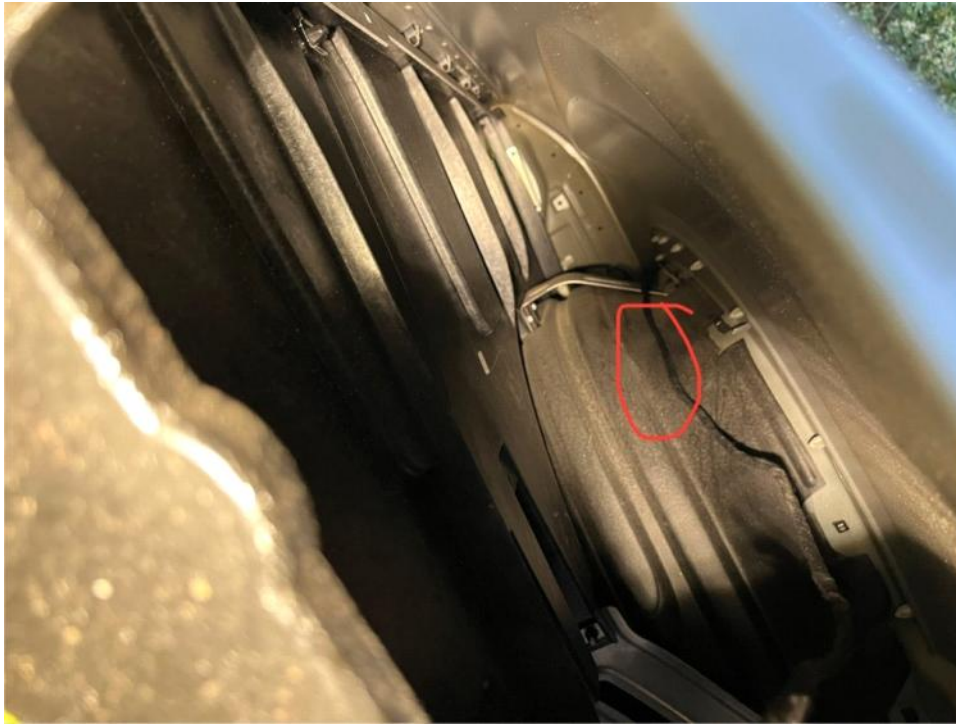
Then I needed something flexible, with some length to it to fish the wire through. My solution? More flags! lol I had this from an old go kart. A broom, extension pole - really anything will work here since it's a large cavity. I taped the other end to the pole and pushed it through.

Just leave some extra around where the tail light opening is as we'll make final connections here later.

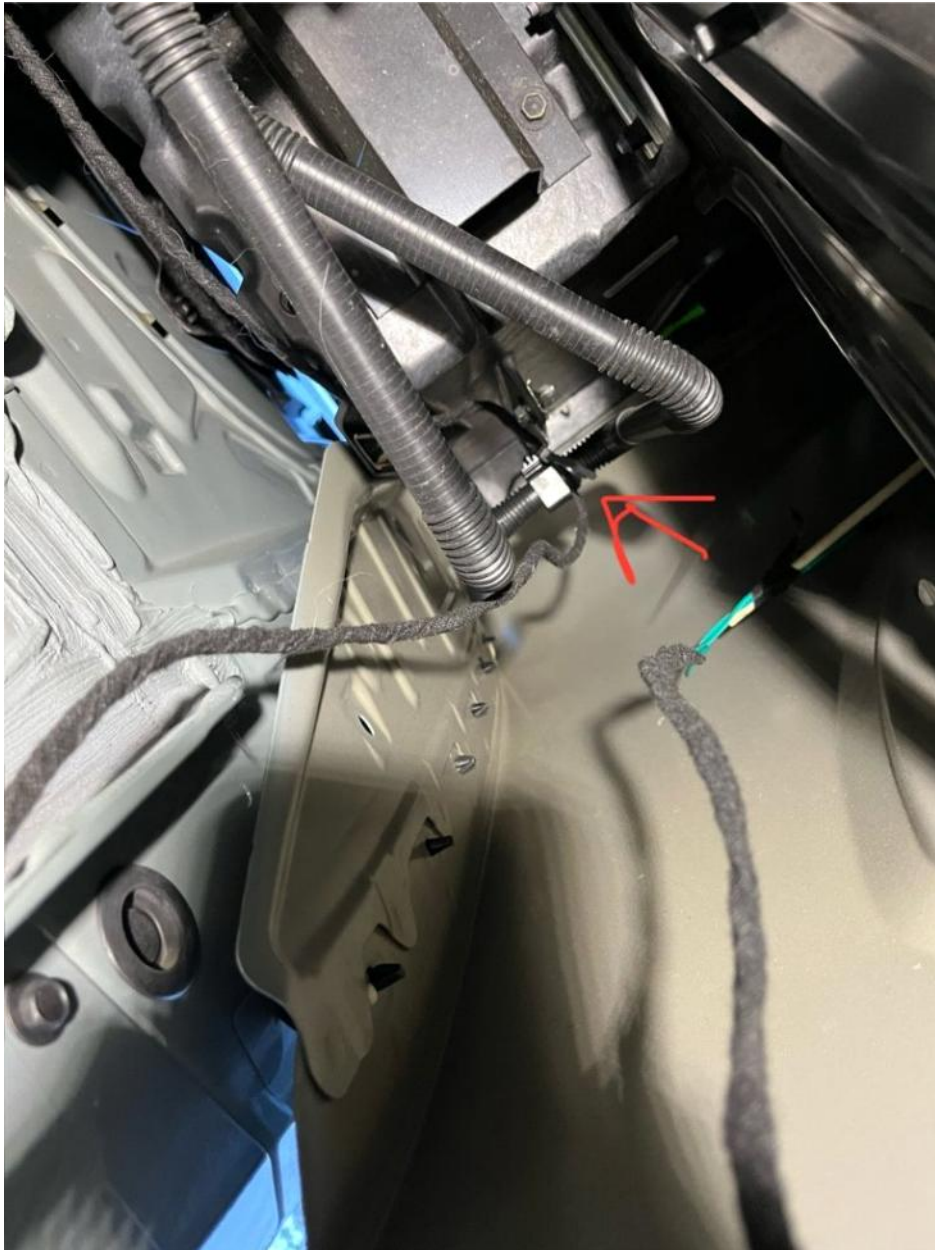




Here's a pic of the wire through the cavity. I got it above that structural member around the wheel well. I circled a section of the wire here so you can see.



Here's the other end of the pole and then I used a zip tie to a bracket for this wire loom where I'm pointing. This is to try to help with support from that long run from the bed location.



After a zip tie goes there to hold it up, I let it drop down following that loom then went on the INSIDE of the frame following the brake lines and did a couple more zip ties (didn't get a picture of this step).

Now this is where the 2 wires will split around halfway down the frame. You can see I have wrapped with the cloth tape up to this split. You can also see, I used the round ground connector that comes in the kit here for one wire. Also notice that one wire is green here - I ended up using some trailer wire I had left over but you can get by with using only what's sent in the kit as he sends plenty - it'll just all be the same black color so will somehow need to keep them identified.



There's a ground point in the top of the frame here. I just backed that bolt out and added my ground wire in line and tightened everything back up.



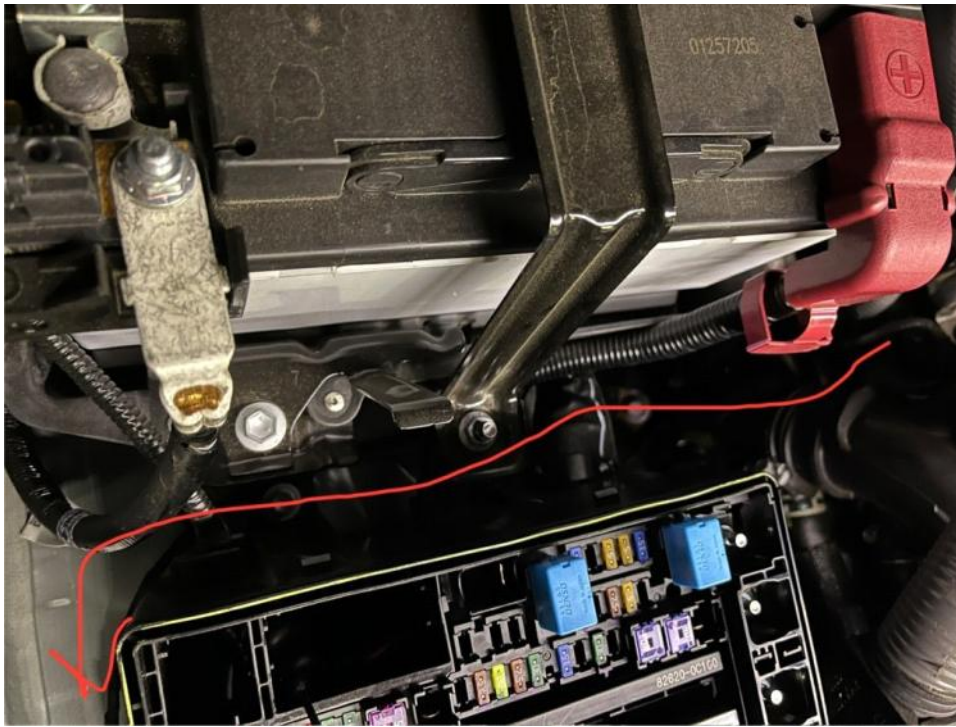
Then I just followed those brake lines up the frame towards the front passenger wheel.



Then I popped the hood and used my go kart flag to fish the wire up towards the fuse box. Here are a couple pics of the location I used.



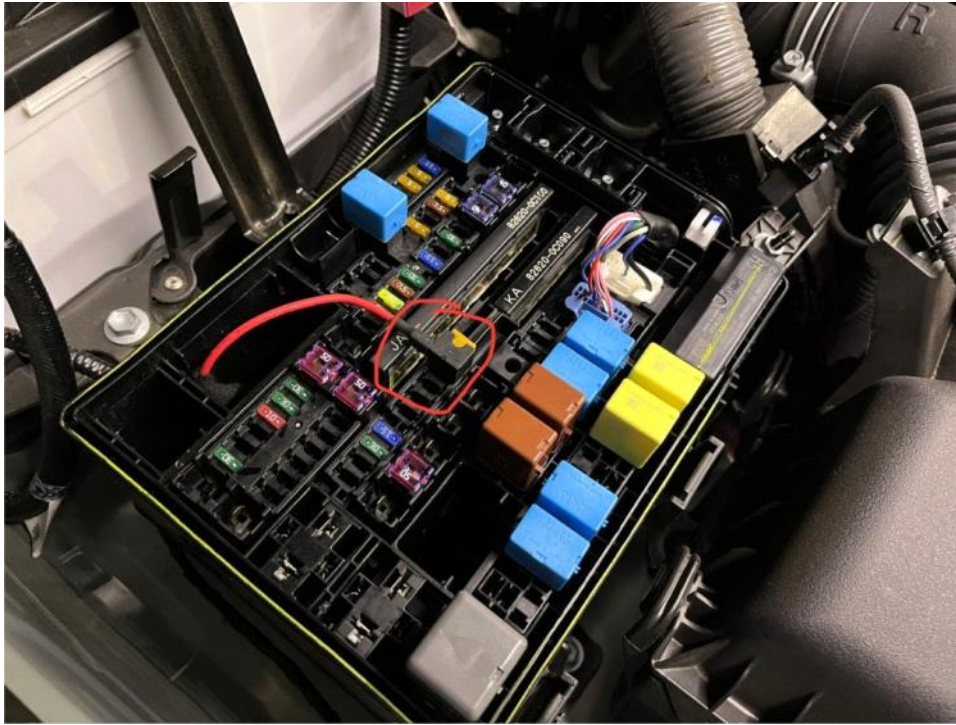
Then I routed it by hand under the large wires between the battery and fuse box. Can't really see the wire here but I have the path drawn in red.



Then I routed around the fuse box and up the entry where the other wires come in.



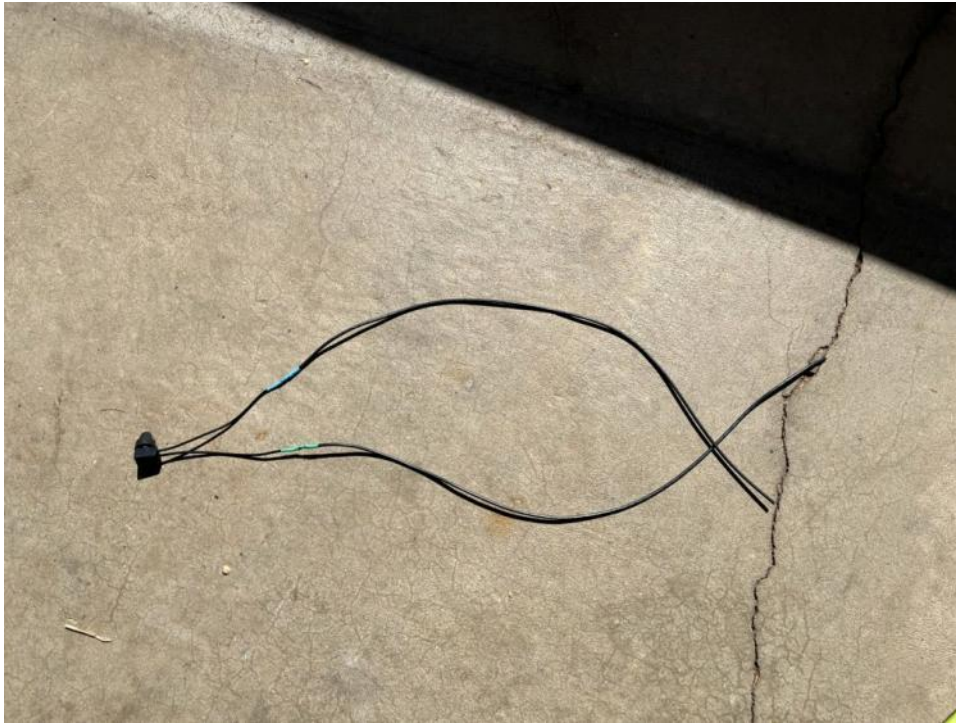
I tested and came up with many constant hot places in the fuse box that weren't occupied but went with this one for ease of installation. I just used the 5A fuse that came with the kit.



Okay, so now you're done with everything upstream. Now to finish the connections downstream where the switch will be for the lights.

There are 2 switches that come with the hood + bed lights kit. I didn't use their switch for the hood lights so had both to choose from. I decided to go with the push button one vs the toggle style. Just personal preference really - you could use either one here.

Here I soldered on a couple longer leads to what came on the push switch then wrapped with my cloth tape.



Then I poked that through the hole where the bed light wires were ran from back earlier in the post.

Here are the final connections. The green going to the red wire in this picture is the run from the +12V at the fuse box directly to the bed lights. Then, the 2 black wires are to the switch. I am switching from ground circuit.





Everything is then wrapped with cloth tape and then zip tied together in a little spool. Then I zip tied this to the small tab that is on a bracket in the well of the tail light area. I did look at this and that's not used by the tail light when it's connected so should be good to use.



Now you should have something like this on the inside of the bed. Now, you can

bolt back on your tail light that you took out - just be careful with the wires behind it not to pinch anything.



Now, how do we secure the switch? I thought about this a good bit. I could have simply drilled a hole in the bed and had this mounted nicely there but my mission is completely reversible and no extra holes. So... I made myself a fancy holder for this on my 3D printer!

I did a quick design where it is mounted to the back of the bed rail. My bed rail is dropped on brackets and has space in the back since I have the top tri-fold cover. I designed it where I can use an M8 bolt with a captured nut on the back end since I couldn't get a tool up there to hold the nut when tightening the bolt from the front end.

I used some transparent PETG. Yes, I probably should have used black but I ran out of it recently and this is all I had. Don't roast me too bad lol







Here it is installed on the rail. It allows other items to slide past it on the rack.



This is the final product! Best pic I got once finished. Everything works great and my mount for the switch is very secure - that PETG material is strong.

